

Velocity-Dependent Dissipative Self-Interacting Dark Matter: A Corrected Gravothermal Fluid Baseline

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Abstract

We revisit the dissipative frequent self-interacting dark matter (dSIDM) framework of [Schmidt et al. \[2026\]](#) by generalizing its velocity-independent transport and cooling terms to Born-limit velocity-dependent elastic cross sections and emission kernels following [Lankester-Broche & Pradler \[2026\]](#). We consider three benchmark scenarios: (i) a dark photon model with massive vector emission ($m_V = 1.1$ keV, $v^* \approx 200$ km/s), (ii) a scalar ϕ model with massive scalar emission ($m_\phi = 6.95$ keV, $v^* \approx 500$ km/s), and (iii) a massless-emission normalization control. The two massive models differ not only in their threshold velocity but also in the velocity dependence of their differential emission kernels. Using a modified version of the GravothermalSIDM fluid code, we define elastic-limit and homogeneous-box regression tests before applying the model to isolated Navarro–Frenk–White (NFW) halos. The present version implements the long-range identical-fermion emission kernels of [Lankester-Broche & Pradler \[2026\]](#) for the χ – V and χ – ϕ channels and uses their energy-weighted integral in the cooling term. Legacy simulation outputs are not treated as quantitative results until they are regenerated with the corrected code. The intended downstream comparison is a multi-model exclusion plot together with a dissipative dimension. Those figures are deliberately deferred until the corrected simulations, absolute B1938 likelihood, and Born-validity mask are in place. The N -body module is omitted from the present analysis and reserved for future validation.

1 Introduction

The standard Λ CDM cosmological model has been spectacularly successful in describing the large-scale structure of the Universe [[Planck Collaboration, 2020](#), [Springel et al., 2006](#)]. However, on galactic and sub-galactic scales, a number of long-standing tensions between theoretical predictions and observations have motivated the exploration of alternatives to collisionless cold dark matter (CDM). These include the core–cusp problem — the predicted cuspy inner density profiles of Λ CDM halos versus the observed cores in dwarf and low-surface-brightness galaxies [[Flores & Primack, 1994](#), [Moore, 1994](#), [de Blok, 2009](#), [Oh et al., 2011](#), [Adams et al., 2014](#)] — the missing-satellites problem — the overabundance of substructure predicted by CDM simulations relative to the observed satellite counts [[Klypin et al., 1999](#), [Moore et al., 1999](#)] — and the diversity of rotation curves in dwarf

galaxies, which span a wide range of inner density slopes difficult to reproduce with a single CDM prescription [Oman et al., 2015, Ren et al., 2019].

Self-interacting dark matter (SIDM), originally proposed by Spergel & Steinhardt [2000], offers a compelling resolution to these small-scale challenges. In the SIDM scenario, dark matter particles scatter elastically through a light mediator, transferring momentum and heat within halos. This process thermalizes the inner regions over a dynamical time, producing observable cores in otherwise cuspy density profiles [Burkert, 2000, Rocha et al., 2013, Tulin & Yu, 2018]. Unlike baryonic feedback mechanisms [Governato et al., 2010, Pontzen & Governato, 2012, Di Cintio et al., 2014], which can also generate cores but with considerable model dependence, SIDM provides a *predictive* and *testable* mechanism: the self-interaction cross section σ/m_χ directly controls the core size and the thermalization timescale, yielding observable correlations between halo properties that are distinct from CDM predictions [Kaplinghat et al., 2016, Creaser et al., 2024].

Observational constraints on the elastic SIDM cross section have tightened considerably over the past decade. The Bullet Cluster [Clowe et al., 2006, Randall et al., 2008] sets an upper bound of $\sigma/m_\chi \lesssim 1\text{--}2\text{ cm}^2/\text{g}$ at cluster velocities ($v \sim 3000\text{ km/s}$), while the observed shapes of cluster cores [Newman et al., 2013, 2015, Harvey et al., 2019] constrain $\sigma/m_\chi \lesssim 0.1\text{--}1\text{ cm}^2/\text{g}$ at $v \sim 1000\text{ km/s}$. At dwarf-galaxy scales ($v \sim 50\text{ km/s}$), the requirement of producing observed cores sets a lower bound of $\sigma/m_\chi \gtrsim 0.1\text{ cm}^2/\text{g}$ [Rocha et al., 2013, Kaplinghat et al., 2015, Ren et al., 2019], while excessive self-interactions would erase too much substructure [Vogelsberger et al., 2012, Zavala et al., 2019]. These multi-scale constraints already suggest that a *velocity-dependent* cross section — decreasing from dwarf to cluster scales — is strongly preferred by the data [Buckley & Fox, 2010, Loeb & Weiner, 2011, Tulin et al., 2013, Kaplinghat et al., 2016].

A natural extension of the SIDM framework introduces *dissipative* self-interactions, in which each scattering event may radiate a dark photon or scalar boson, removing energy from the dark sector. Such dissipative SIDM (dSIDM) models arise in many well-motivated dark sector constructions, including atomic dark matter [Kaplan et al., 2010, Cyr-Racine & Sigurdson, 2014], mirror dark matter [Foot, 2014, Foot & Vagnozzi, 2014], self-interacting hidden sector models [Essig et al., 2019, Berlin et al., 2018], and models with dark radiation emission [Chacko et al., 2018, Berlin & Tulin, 2019]. The dissipative channel introduces a new dimension beyond the elastic cross section: the dissipation parameter $r_{\text{diss}} \geq 1$, which quantifies the ratio of inelastic to elastic energy loss per collision. Even weak dissipation ($r_{\text{diss}} - 1 \sim 0.01\text{--}0.05$) can qualitatively alter halo evolution, accelerating core collapse and producing more compact substructure than purely elastic SIDM [Schmidt et al., 2026].

The recent work of Schmidt et al. [2026] developed the first gravothermal fluid treatment of *frequent* small-angle SIDM with dissipation (dubbed dSIDM/fSIDM), demonstrating that dissipative models can explain the compact substructure observed in the strong-lensing system JVAS B1938+666 [Vegetti et al., 2026] with significantly smaller cross sections than purely elastic SIDM. The B1938+666 system is particularly valuable as a probe of dark matter substructure: the strong-lensing perturbation provides two mass measurements at projected radii of 20 and 90 pc, constraining the projected mass ratio to $M(20\text{ pc})/M(90\text{ pc}) = 0.364 \pm 0.022$ at a lens redshift $z = 0.881$ [Vegetti et al., 2012, 2026, Hezaveh et al., 2016].

However, Schmidt et al. [2026] employed *velocity-independent* σ_T/m_χ and r_{diss} , which preserves a scaling symmetry (Appendix G of their paper) that prevents translating halo-scale observations into particle-physics parameters. This symmetry arises because, for

constant cross sections, the dimensionless scattering number $n_\sigma = \sigma \rho v t$ is invariant under a simultaneous rescaling of length, mass, and time. To break this degeneracy and connect astrophysical measurements to dark sector Lagrangians, one must consider *velocity-dependent* cross sections derived from specific particle-physics models, as the velocity dependence introduces a physical scale that is not preserved by the rescaling symmetry.

In this work, we establish the implementation needed for that next step. Using the Born-limit transfer cross section and the identical-fermion energy-differential kernels of [Lankester-Broche & Pradler \[2026\]](#), we construct a direct cooling closure for three benchmark dark sector models: (i) a dark photon model with massive vector emission, (ii) a scalar ϕ model with massive scalar emission, and (iii) a massless control. The M1 and M2 kernels are evaluated numerically over emitted-particle energy and then averaged over the relative-speed Maxwell distribution. The older $r_{\text{diss}}(v)$ threshold ansatz is retained only for diagnostics; it is no longer used to define the preferred cooling rate.

Our immediate objective is to establish a validated route from these velocity-dependent closures to dwarf-, lens-, and cluster-scale observables. Model discrimination is therefore treated as a future output of the corrected simulations, rather than as a result of the legacy exclusion curves. The final comparison must also propagate observational likelihoods and the domain of validity of the Born approximation.

The GADGET-4 N -body implementation is retained in the repository as unfinished infrastructure, but is not used as a validation result here: its small-angle fSIDM correspondence and resolution convergence have not yet been established. Independent particle validation is deferred until the fluid cooling law and its microphysical normalization are fixed.

The paper is organized as follows. Section 2 describes the dark sector models and the velocity-dependent cross sections, including the correction factors C_V and C_ϕ . Section 3 derives the dissipative cooling rate used in the fluid code and discusses its relation to the volumetric rate of [Schmidt et al. \[2026\]](#). Section 4 presents the fluid simulations and validation. Section 5 applies the B1938+666 rescaling and defines the validation protocol. The exclusion-plot framework is described as a future analysis target; the present version does not treat the legacy curves as results. Section 7 documents the limitations and caveats. We conclude in Section 8.

2 Dark sector models and velocity-dependent cross sections

2.1 Unified Born framework

We use the unified Born-limit framework of [Lankester-Broche & Pradler \[2026\]](#) as a source of model-motivated velocity dependence. The transfer cross section is implemented explicitly, whereas the dissipative rate is obtained from the energy-differential kernels for the supported long-range identical-fermion channels. For identical-particle self-scattering (the case relevant for halo dynamics), the dipole-emission channel vanishes and the quadrupole channel must be treated consistently.

The transfer cross section in the Born limit is

$$\sigma_T(v_{\text{rel}}) = \frac{a^4}{8\pi\mu^2 v_{\text{rel}}^4} \left[\ln \left(1 + \frac{4\mu^2 v_{\text{rel}}^2}{m^2} \right) - \frac{4\mu^2 v_{\text{rel}}^2/m^2}{1 + 4\mu^2 v_{\text{rel}}^2/m^2} \right], \quad (1)$$

where $\mu = m_\chi/2$ is the reduced mass, m is the mediator mass, v_{rel} is the relative speed, and $a = \sqrt{4\pi\alpha_D}$ for the vector benchmark (with the analogous effective coupling for the scalar benchmarks). For $x = 4\mu^2 v_{\text{rel}}^2/m^2 \ll 1$, the bracket is $x^2/2$ and the cross section approaches a constant. For $x \gg 1$, it scales as $v_{\text{rel}}^{-4}[\ln x - 1]$. These limits are used below when checking the validity of the Born approximation.

2.2 Legacy dissipation diagnostic

Before the direct kernel was implemented, the project modeled massive emission through a velocity-dependent phenomenological dissipation parameter. We retain this ansatz only to diagnose legacy outputs; it is not used by the preferred M1/M2 cooling path. The relevant threshold velocity is

$$v^* = \sqrt{\frac{2m_{\text{med}}}{\mu}}. \quad (2)$$

The legacy closure is

$$r_{\text{diss}}(v_{\text{rel}}) = 1 + C_0 C_{V/\phi}(\omega) \exp\left(-\frac{m_{\text{med}}}{\frac{1}{2}\mu v_{\text{rel}}^2/c^2}\right), \quad (3)$$

where $C_0 = 0.05$ is an explicit fiducial phenomenological amplitude, not a derived prediction. This expression is retained only as a threshold diagnostic; quantitative cooling for the supported channels uses the full energy integral described in Section 3.

Model-dependent pre-exponential factor. Within this legacy ansatz, the two massive benchmarks differ not only in their threshold velocity v^* , but also in the *shape* of $r_{\text{diss}}(v)$ near threshold, via the correction factors derived by [Lankester-Broche & Pradler \[2026\]](#):

$$C_V(\omega) = 1 + \frac{m_V^2}{2\omega^2} \quad (\text{vector; LB2026 Eq. 3.27; enhanced near threshold}), \quad (4)$$

$$C_\phi(\omega) = \max\left(0, 1 - \frac{m_\phi^2}{\omega^2}\right) \quad (\text{scalar; LB2026 Eq. 3.26; suppressed near threshold}), \quad (5)$$

where $\omega \sim \frac{1}{2}\mu v_{\text{rel}}^2$ is the characteristic emitted energy (the CM kinetic energy per particle). The physical origin is the longitudinal polarization of the massive vector (which adds an extra radiated mode near threshold), versus the kinematic phase-space closure of the scalar emission amplitude (which vanishes at $\omega = m_\phi$). We evaluate $C_{V/\phi}$ at $\omega = \frac{1}{2}\mu v_{\text{rel}}^2/c^2$ as the characteristic scale.

For $v_{\text{rel}} \gg v^*$, this diagnostic tends to constant r_{diss} ; for $v_{\text{rel}} \ll v^*$ it tends to the elastic limit. These asymptotes are useful regression checks, but model comparisons use the direct differential kernels, not the intermediate behavior of Eq. (3).

2.3 Benchmark models

We consider three benchmarks with $m_\chi = 10$ GeV:

The M1 and M2 thresholds lie near dwarf and cluster velocity scales, respectively. M3 is a massless-emission normalization control; its elastic Rutherford cross section is regulated in code with a 10 eV mediator mass and does not represent the constant-cross-section Schmidt benchmark.

Table 1: Benchmark dark sector models.

Model	Type	m_{em} [keV]	v^* [km/s]	α_D	Emission
M1	Dark photon (vector)	1.1	200	10^{-4}	Massive
M2	Scalar ϕ	6.95	500	10^{-4}	Massive
M3	Control (vector)	0	—	10^{-4}	Massless

These fiducial points are retained only to document the legacy setup. The Yukawa Born expansion parameter $\eta_B = \alpha_D \mu / m$ is approximately 455, 72, and 5×10^4 for M1, M2, and the M3 elastic regulator, respectively, and hence is not small. None of the three fiducial points is a controlled Born-limit benchmark. The parameter scan now masks points with $\eta_B \geq 1$; a stricter threshold is required for precision predictions.

Born-valid parameter audit. At the boundary $\eta_B = 1$, the largest available transfer cross sections at 100 km/s are only 3.23×10^{-4} , 9.72×10^{-3} , and 4.33×10^{-8} cm²/g for M1, M2, and the M3 regulator, respectively. We therefore cannot simultaneously impose the legacy 50 cm²/g calibration and remain in the Born regime. The machine-readable scan samples $\eta_B = 0.01$ – 0.99 and marks $\eta_B \leq 0.1$ as the controlled regression subset. At fixed M1/M2 threshold velocities, varying the mass hierarchy changes this conclusion: the $\eta_B = 0.1$ scan reaches 0.1–10 cm²/g at 100 km/s for approximately $m_\chi = 0.10$ – 0.32 GeV (M1) and 0.22–0.89 GeV (M2). These ranges motivate the controlled representative points used below.

3 Dissipative cooling rate

Schmidt et al. [2026] give the volumetric cooling rate as (their Eq. 19)

$$C_{\text{vol}}(\rho, \nu) = \frac{8}{\sqrt{\pi}} \frac{\sigma_T}{m_\chi} (r_{\text{diss}} - 1) \rho^2 \nu^3, \quad (6)$$

where ν is the 1D velocity dispersion and ρ is the mass density. Our GravothermalSIDM implementation uses a per-unit-mass form, $C_{\text{cool}} = C_{\text{vol}}/\rho$.

$$C_{\text{cool}} = \alpha \frac{\sigma_T}{m_\chi} \rho \nu^3 (r_{\text{diss}} - 1), \quad (7)$$

with the coefficient fixed to $\alpha = 8/\sqrt{\pi}$ by the source equation. This is the constant-cross-section limit. For a velocity-dependent emission kernel, define

$$Q(v) = \int_{m_{\text{em}}}^{\mu v^2/2} d\omega \omega \frac{d\sigma}{d\omega} \quad (8)$$

as the emitted-energy-weighted cross section at relative speed v . Matching the binary loss rate to the fluid convention gives

$$\left(\frac{\sigma}{m_\chi} \right)_{\text{cool}}(v) = \frac{2Q(v)}{m_\chi^2 v^2}, \quad C_{\text{vol}} = \frac{\rho^2}{4} \left\langle \left(\frac{\sigma}{m_\chi} \right)_{\text{cool}}(v) v^3 \right\rangle_{\text{rel}}. \quad (9)$$

The factor 1/4 is the identical-particle pair factor. This expression reduces exactly to Eq. (6) for constant σ_T/m_χ and constant r_{diss} .

Implementation status. The code uses the source coefficient directly and updates the specific internal energy with $-\Delta t C_{\text{cool}}$. For M1 and M2, it evaluates Eqs. (3.43) and (3.47) of Lankester-Broche & Pradler [2026], integrates $Q(v)$ over emitted energy, and performs the relative-speed average in Eq. (9).

Normalization status. The previous $1/6$ coefficient was a calibration constant and has been removed. Quantitative dissipative runs must be regenerated after this correction; no elastic core-collapse time is used to normalize cooling, because the cooling term vanishes in the elastic limit. The implemented kernel currently covers only the long-range identical-fermion χ - V and χ - ϕ channels; scalar and short-range channels remain future work.

4 Fluid simulations

4.1 GravothermalSIDM with dissipation

We modified the public GravothermalSIDM code¹ to accept velocity-dependent $\sigma_T(v_{\text{rel}})$ and $r_{\text{diss}}(v_{\text{rel}})$. The key modifications are:

1. Pairwise relative velocities are Maxwell distributed. For identical populations, each component has variance $2\nu^2$.
2. The energy equation includes the source cooling law and, for M1/M2, the energy-weighted moment in Eq. (9).
3. An additional timestep criterion based on the cooling timescale.

The initial NFW profile uses $r_s = 3.6$ kpc and $\rho_0 = 7.09 \times 10^{-3} M_{\odot} \text{pc}^{-3}$. The legacy setup used $\sigma_T/m_{\chi} = 50 \text{ cm}^2/\text{g}$ at $v_{\text{ref}} = 100 \text{ km/s}$; the Born-valid regressions instead use the cross section implied by their microphysical coupling.

4.2 Validation status

The numerical values in the previous validation subsection were generated before the cooling-rate and relative-speed corrections described above. They are retained only as a record of the legacy run and are excluded from the present analysis. New elastic and dissipative regression runs are required before quoting collapse times or model rankings. As an interface test, we evolved 48-shell M1 and M2 halos for 25 steps at $\eta_B = 0.1$. Over approximately 10^{-3} Gyr, the central-density changes were -1.18×10^{-6} and -2.43×10^{-5} , respectively. These short runs verify numerical stability and the expected weak-evolution limit; they are not production collapse histories. The mass-hierarchy scan also supplies $\eta_B = 0.1$ points with $\sigma_T/m_{\chi} = 1 \text{ cm}^2/\text{g}$ at 100 km/s: $m_{\chi} = 0.148 \text{ GeV}$, $m_V = 16.3 \text{ eV}$ for M1 and $m_{\chi} = 0.460 \text{ GeV}$, $m_{\phi} = 0.320 \text{ keV}$ for M2. In 25-step paired runs, the elastic-only and direct-cooling trajectories agree because the initial cooling moment is below 1.4×10^{-36} in code units: the halo is below the massive-emission threshold. This is a threshold-regime check, not evidence for a B1938 fit.

¹<https://github.com/kboddy/GravothermalSIDM>

5 B1938+666 rescaling

5.1 Method

The strong-lensing perturbation in JVAS B1938+666 [Vegetti et al., 2026] provides two mass measurements: $M(r < 20 \text{ pc}) = (4.25 \pm 0.21) \times 10^5 M_\odot$ and $M(r < 90 \text{ pc}) = (1.167 \pm 0.039) \times 10^6 M_\odot$, at lens redshift $z_{\text{obs}} = 0.881$ (corresponding to $t_{\text{zobs}} = 6.37 \text{ Gyr}$). Following Appendix G of Schmidt et al. [2026], we scan the projected radius r_{2D}/r_s for each snapshot to find where the dimensionless mass ratio $M(r < r_{2D})/M(r < 4.5 r_{2D})$ matches the observed 0.364 ± 0.022 within 3σ . This ratio-only filter is an exploratory preselection, not a likelihood: the two absolute projected masses are therefore evaluated jointly below. For each match we compute the rescaling parameters

$$\lambda = \frac{20 \text{ pc}}{(r_{2D}/r_s) r_{s,\text{sim}} 1000 \text{ pc/kpc}}, \quad (10)$$

$$\mu = \frac{M_{\text{sim}}^{\text{in}} M_{\text{obs}}^{\text{in}} / \sigma_{\text{in}}^2 + M_{\text{sim}}^{\text{out}} M_{\text{obs}}^{\text{out}} / \sigma_{\text{out}}^2}{(M_{\text{sim}}^{\text{in}})^2 / \sigma_{\text{in}}^2 + (M_{\text{sim}}^{\text{out}})^2 / \sigma_{\text{out}}^2}, \quad (11)$$

and the physical parameters $r_s^{\text{phys}} = \lambda r_s^{\text{sim}}$, $\rho_0^{\text{phys}} = \mu \rho_0^{\text{sim}} / \lambda^3$, $\sigma_T / m_\chi^{\text{phys}} = (\lambda^2 / \mu) \sigma_T / m_\chi^{\text{sim}}$, $t_{\text{evo}} = \sqrt{\lambda^3 / \mu} t_{\text{sim}}$. The two absolute masses are evaluated with

$$\chi_M^2(\mu) = \sum_{i \in \{20, 90 \text{ pc}\}} \left[\frac{\mu M_i^{\text{sim}} - M_i^{\text{obs}}}{\sigma_i} \right]^2, \quad (12)$$

whose minimum gives the estimator for μ above. The implementation records both standardized residuals and χ_M^2 for every ratio-preselected point.

Important caveat: rescaling validity. The above formulas are exact only for the elastic, velocity-independent equations. They are used here only to construct initial guesses for a future direct physical-parameter scan. No model-dependent B1938 band is claimed until the corrected fluid equations are rerun at the physical velocity scale and both absolute mass measurements are fitted.

5.2 Results status

The legacy B1938 tables and resimulation offsets are disabled because they were generated with the pre-correction cooling law and a ratio-only scan. The joint absolute-mass likelihood is now implemented and was exercised on the short $\eta_B = 0.1$ regressions. The best evolved diagnostic points have $\chi_M^2 \simeq 0.051$ for one fitted degree of freedom, but their rescaled age is only $1.2 \times 10^{-5} \text{ Gyr}$ and their profiles remain essentially initial NFW profiles. They validate the likelihood interface only. A physical B1938 result still requires long corrected runs, with the compact and extended solution branches and concentration prior treated separately.

6 Exclusion-plot status

The previous exclusion figures are disabled because they combine hand-set rectangular bounds with model curves that depend on the pre-correction thermal average and dissipative normalization. The corrected analysis will need velocity-distributed likelihoods and a stated validity mask for the Born approximation before any exclusion claim is restored.

7 Limitations and caveats

We document the following limitations of the present analysis, in decreasing order of importance:

1. The corrected cooling implementation has not yet been propagated through the production fluid runs. All legacy collapse times, rescaling offsets, grid counts, and exclusion curves are therefore provisional and are not used as quantitative conclusions.
2. The implemented direct emission kernel covers only the long-range identical-fermion χ - V and χ - ϕ channels. Scalar and short-range kernels, together with their complete interference terms, still require implementation before a general model scan is possible.
3. The Born-valid region does not overlap the legacy $50 \text{ cm}^2/\text{g}$ calibration. At $\eta_B < 1$ the M1/M2 cross sections at 100 km/s remain below approximately 3.23×10^{-4} and $9.72 \times 10^{-3} \text{ cm}^2/\text{g}$. A fixed-threshold mass-hierarchy scan supplies controlled candidates at lower masses, but their long-time halo evolution remains uncomputed.
4. The B1938 joint absolute-mass likelihood is implemented, but a physical inference still requires long direct runs, the extended solution branch, and a concentration prior; the ratio grid remains only a preselection.
5. The N-body module is excluded from the present paper and remains future validation work.

8 Discussion and conclusion

The present paper establishes a corrected implementation baseline rather than a final exclusion result. The fluid code now uses the source cooling coefficient, the required ν^3 energy-rate scaling, and the relative-speed Maxwell distribution for identical particles. These changes invalidate the legacy production outputs, so no collapse time, B1938 matching count, or model-exclusion claim is reported here.

The next quantitative release must evolve the mass-hierarchy candidates long enough to test whether their Born-valid cross sections produce the required halo structure. If they do not, the project must replace the Born calculation with a controlled nonperturbative treatment. It must then extend the direct kernel to the remaining channels, regenerate the fluid runs, and fit both absolute B1938 projected masses. The N-body module is intentionally excluded from the present conclusions and remains future validation work.

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